

Preliminary Empirical Results

Testing the Zoning–Transportation Mechanism

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Model: Setup

Metro area S partitioned into municipalities $g \in G$; fixed total population N .

Households

Cobb-Douglas over consumption & housing;
Fréchet taste shocks $v_{ij}(\omega)$.

Choose (i, j) ; earn w_j , pay q_i^R , bear
congestion disutility

$$\phi(N_i, L_i) = \eta_N N_i^{1+\rho} + \eta_L L_i^{1+\rho}$$

Location shares: $\pi_{ij} \propto T_i E_j \tilde{V}_{ij}^\epsilon$

Firms at j

CRS production; commercial land $\bar{H}_j^F$
set by zoning.

Governments

Municipality g : sets $\bar{H}_i^R$ and $\bar{H}_i^F$

Transport authority: sets $\{I_{jk}\}$

$\Rightarrow$ Simultaneous Nash equilibrium

Zoning proxy: housing supply elasticity $\hat{\gamma}_i$ from Baum-Snow and Han (2024) (lower $\hat{\gamma}_i$
= tighter zoning = smaller $\bar{H}_i^R$).

Municipal Problem: FOC $\Rightarrow$ Propositions 1 & 2

Municipality g maximizes $W^g = \sum_{i \in S^g} N_i (\bar{U}_i - \phi(N_i, L_i))$.

Interior FOC for residential capacity $\bar{H}_i^R$:

$$\underbrace{\bar{U}_i \frac{\partial N_i}{\partial \bar{H}_i^R}}_{\substack{(+)\text{ in-movers}}} + \underbrace{N_i \frac{\partial \bar{U}_i}{\partial q_i^R} \frac{\partial q_i^R}{\partial \bar{H}_i^R}}_{\substack{(+)\text{ rent falls}}} + \underbrace{N_i \frac{\partial \bar{U}_i}{\partial \phi} \frac{\partial \phi}{\partial N_i} \frac{\partial N_i}{\partial \bar{H}_i^R}}_{\substack{(-)\text{ congestion cost}}} + \underbrace{N_i \left(-\frac{\partial \bar{U}_i}{\partial \phi} \frac{\partial \phi}{\partial L_i} \frac{\partial L_i}{\partial \bar{H}_i^R} \right)}_{\substack{(+)\text{ less worker influx}}} = 0$$

Proposition 1

$\partial \phi / \partial L_i = (1 + \rho) \eta_L L_i^\rho$ increases in L_i .

Higher $L_i \Rightarrow$ larger *worker-influx* cost in FOC
 $\Rightarrow$ municipality restricts residential zoning.

Test: $\partial \hat{\gamma}_i / \partial \log L_i < 0$

Proposition 2

Transit $\Rightarrow$ higher $\tilde{V}_{ij} \Rightarrow$ upward pressure on N_i .

Larger N_i amplifies the *congestion* term $\Rightarrow$ municipality preemptively tightens zoning.

Test: $\hat{\gamma}_i|_{\text{near transit}} < \hat{\gamma}_i|_{\text{far}}$

Transportation Authority: FOC $\Rightarrow$ Proposition 3

Authority maximizes $N \cdot \Gamma(1 - \frac{1}{\epsilon}) \Phi^{1/\epsilon}$ subject to $\sum_{jk} \delta_{jk} I_{jk} \leq K$.

Interior FOC for link (j, k) :

$$\underbrace{\frac{N}{\epsilon} \Gamma(1 - \frac{1}{\epsilon}) \Phi^{1/\epsilon-1} \frac{\partial \Phi}{\partial I_{jk}}}_{\text{Marginal Social Benefit (MSB)}} = \mu \delta_{jk}$$

The MSB increases in ridership on link (j, k) :

$$\frac{\partial \Phi}{\partial I_{jk}} \propto \underbrace{\sum_{\text{users of } (j,k)} n_{ij}}_{\text{ridership}} \times \underbrace{\left| \frac{\partial \tau_{ij}}{\partial I_{jk}} \right|}_{\text{cost reduction}}$$

Transit ridership n_{ij} depends on the density of residents *and* jobs near the corridor.

Proposition 3

$$\underbrace{\text{Restrictive zoning}}_{\tilde{H}_i^R \downarrow} \Rightarrow \underbrace{N_i \downarrow, L_j \downarrow}_{\text{near transit}} \Rightarrow \underbrace{n_{ij} \downarrow}_{\text{ridership}} \Rightarrow \underbrace{\partial \Phi / \partial I_{jk} \downarrow}_{\text{MSB}} \Rightarrow I_{jk}^* \downarrow$$

Test: station-area density/ridership response to transit is weaker where land-use capacity is lower.

Overview: Three Proposition Tests

Empirical chain from the model's FOC

- ❶ **P1.** High employment concentration $\Rightarrow$ tighter residential zoning
- ❷ **P2.** Transit access raises population pressure $\Rightarrow$ tighter zoning response
- ❸ **P3.** Restrictive land use $\Rightarrow$ lower station-area transit returns

Data

- Tract-level housing supply elasticity $\hat{\gamma}_i$: Baum-Snow and Han (2024), Dallas metro ($N = 1,016$ tracts)
- Employment inflows L_i : LODES 2023, aggregated to Census tract
- Transit access: DART station locations (148 stations), 0.5-mile buffer
- P3 data audit: existing NHGIS extract has tract population and housing units, but is not yet sufficient for a station-buffer panel

Tests are equilibrium consistency checks, not causal estimates. Each proposition maps to a model FOC.

P1: Specification

Model prediction (from municipal FOC, worker-influx term):

$$\frac{\partial \hat{\gamma}_i}{\partial \log L_i} < 0$$

Higher employment at a tract raises the marginal disutility of additional workers, so municipalities choose tighter residential zoning.

OLS regression (Census tract, Dallas metro; $N = 1,016$):

$$\hat{\gamma}_i = \alpha + \underbrace{\beta_1}_{< 0} \log(1 + L_i) + \beta_2 \text{pctdis}_i + \varepsilon_i$$

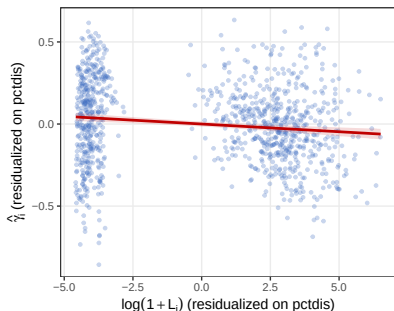
Variable	Definition
$\hat{\gamma}_i$	Housing supply elasticity (Baum-Snow & Han 2024); <i>lower = tighter zoning</i> (proxies $\bar{H}_i^R$ in the model)
$\log(1 + L_i)$	Log employment inflows at tract i from LODES 2023 (proxies L_i in the model)
pctdis_i	Fractional distance from tract centroid to CBD; controls for the

P1: Employment Concentration and Zoning Restrictiveness

Prediction. From the municipal FOC,

$$\frac{\partial \hat{\gamma}_i}{\partial \log L_i} < 0.$$

High-employment tracts face larger marginal congestion costs from looser zoning.



Residualized on CBD distance ($\widehat{\text{pctdis}}$).

	(1) $\hat{\gamma}_{01a}$	(2) $\hat{\gamma}_{01a}$	(3) $\hat{\gamma}_{11a}$	(4) $\hat{\gamma}_{01a,sp}$
$\log(1 + L_i)$	-0.009*** (0.002)	-0.009*** (0.002)	-0.006*** (0.002)	-0.009*** (0.002)
pctdis_i		-0.087* (0.045)	-0.095** (0.045)	-0.040 (0.042)
R^2	0.016	0.020	0.011	0.017
N	1,016	1,016	1,016	1,016

Result

Negative and significant across all four specifications. Consistent with the municipal congestion-cost FOC.

P2: Specification

Model prediction (from municipal FOC, congestion term):

$$\hat{\gamma}_i|_{\text{near transit}} < \hat{\gamma}_i|_{\text{far}}$$

Transit access raises residential attractiveness $\tilde{V}_{ij}$, increasing N_i pressure; municipalities respond by preemptively tightening zoning.

OLS regression (Census tract, Dallas metro; $N = 581$):

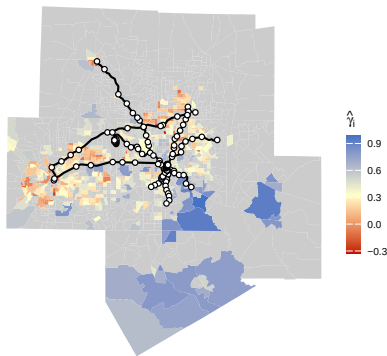
$$\hat{\gamma}_i = \alpha + \underbrace{\beta_1}_{< 0} \mathbf{1}[\text{near DART}]_i + \beta_2 \text{pctdis}_i + \beta_3 \log(1 + L_i) + \varepsilon_i$$

Variable	Definition
$\hat{\gamma}_i$	Housing supply elasticity; <i>lower = tighter zoning</i>
$\mathbf{1}[\text{near DART}]_i$	= 1 if tract centroid within 0.5 mi of a DART rail station (148 stations in DFW)
pctdis_i	Fractional distance to CBD (same as P1)
$\log(1 + L_i)$	Log employment inflows (same as P1; added as control)

P2: DART Access and the Zoning-Tightening Response

Prediction. Transit access raises local attractiveness; municipalities respond by tightening residential capacity:

$$\hat{\gamma}_i|_{\text{near DART}} < \hat{\gamma}_i|_{\text{far}}.$$



Blue = high $\hat{\gamma}_i$ (permissive); red = low (restrictive).

	(1)	(2)	(3)	(4)
1[near DART]	-0.152** (0.059)	-0.205*** (0.058)	-0.178*** (0.057)	-0.177** (0.079)
pctdis		-0.412*** (0.059)	-0.384*** (0.059)	-0.384*** (0.059)
log(1 + L_i)			-0.032*** (0.007)	-0.032*** (0.007)
1[near] $\times$ pctdis				-0.011 (0.418)
R^2	0.011	0.087	0.118	0.118
N	581	581	581	581

N = 581 Baum-Snow tracts matched to 2020 Census

geometry; 15 have DART-adjacent centroid.

Result

Near-DART tracts have elasticities 0.18 units lower. Interaction insignificant (few suburban near-station tracts).

P3: NHGIS Data Audit

Question. Can the NHGIS files currently in /Volumes/ORICO/Proposal for SYP support the requested station-area P3 test?

What is available

- Two tract-level NHGIS nominal time-series CSV extracts
- DFW county coverage: 2,856 integrated tract rows
- Key available measures: total population, total housing units, and travel-time-to-work bins
- Census years include 1990, 2000, 2010, and 2020

Why this is not sufficient

- No NHGIS GIS boundary shapefiles for 1990/2000/2010/2020
- Nominal tracts are not harmonized geographies; codes can refer to changed areas over time
- No block-group extract, limiting 0.5-mile station-buffer precision
- Missing controls/outcomes: income, tenure, vacancy, race/ethnicity, commute mode or public-transit share

Conclusion

The existing NHGIS data can help seed a tract-level panel, but it is not enough to estimate P3 as specified in the project instructions.

P3: What to Download from NHGIS

Download a DFW-focused NHGIS extract at tract or, preferably, block-group geography.

Need	NHGIS request
Years	1990, 2000, 2010, 2020; add ACS 2016–2020 and 2020–2024 if using commuting, income, or tenure controls
Geography	Block groups preferred; tracts acceptable for first version. Use geographically standardized 2010 or 2020 units if available
GIS files	Boundary shapefiles for every source year and the chosen target year, or NHGIS standardized GIS files if using standardized tables
Core outcomes	Total population, total housing units, occupied units, vacant units, owner/renter occupied units
Controls	Median household income, race/ethnicity counts, land area if available; otherwise compute area from GIS in EPSG:2276
Transit response	Means of transportation to work, especially public transportation/transit commute counts or share; mean commute time if available

Storage convention

Place raw NHGIS tables and shapefiles in `data/raw/nhgis/`. If geographically standardized files are unavailable, also add crosswalks to `data/raw/census_crosswalks/`; otherwise use area-weighted overlay.

P3: Revised Station-Area Specification

Model prediction (from transportation authority FOC): restrictive land use lowers the density/ridership response that enters transit's marginal social benefit.

$$Y_{sbt} = \alpha_s + \lambda_t + \beta_1 PostTransit_{st} + \underbrace{\beta_2}_{< 0} (PostTransit_{st} \times TightZoning_b) + X'_{bt} \Gamma + \varepsilon_{sbt}$$

Variable	Definition
s, b, t	Station, intersected buffer geography, and year
Y_{sbt}	Population density, housing-unit density, transit commute share, or station ridership if available
$PostTransit_{st}$	Indicator for years after station s opens
$TightZoning_b$	Direct zoning capacity, minimum lot size, single-family share, or low pre-opening density/built-form proxy
X_{bt}	Income, race/ethnicity, tenure, vacancy, and CBD-distance controls

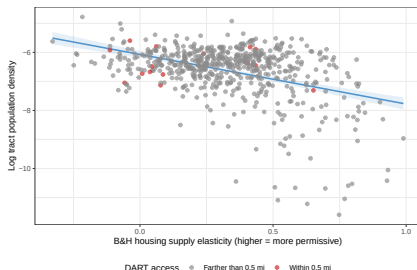
Interpretation

This is the empirical object in the instructions: realized local development or ridership response to transit, not a cross-metro network-length regression.

P3: Interim Evidence from Existing Files

The full event-study panel cannot be run until the NHGIS GIS and additional attributes are added. As an interim check, I use the existing 2020 tract data:

$$\log(\text{density}_i) = \alpha + \beta_1 \text{Tight}_i + \beta_2 \text{NearDART}_i + \beta_3 (\text{Tight}_i \times \text{NearDART}_i) + X_i' \Gamma + \varepsilon_i.$$



Dallas matched tracts: N = 593; near-DART tracts = 15

	Log pop. density	Log unit density
Tightness $\times$ near DART	-1.747** (0.786)	-1.507* (0.781)
Near DART, 0.5 mi	-0.923*** (0.231)	-0.841*** (0.229)
Controls	Yes	Yes
R^2	0.488	0.512
N	590	590

Read cautiously

Around DART, tighter B&H supply elasticity is associated with a weaker density base. This is consistent with the P3 mechanism, but it is a cross-section and cannot yet estimate post-opening returns.

Interpretation and Next Steps

What the tests show

- **P1** ✓ Employment-concentrated tracts are systematically less permissive, consistent with the municipal congestion-cost FOC.
- **P2** ✓ DART-adjacent tracts are significantly less permissive than matched non-adjacent tracts, consistent with the transit-tightening channel.
- **P3 data gap** Current NHGIS files are not sufficient for the requested station-buffer event-study panel.
- **P3 interim** Existing 2020 tract evidence is consistent with weaker density near DART where land-use capacity is tighter, but it is not yet a causal return estimate.

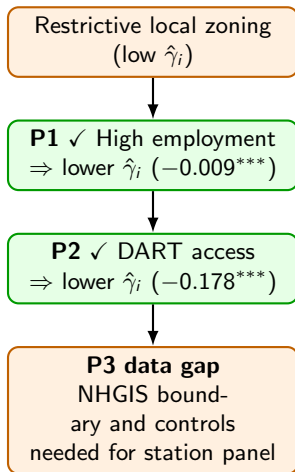
What's actually needed for P3

- NHGIS tract or block-group GIS boundaries for 1990, 2000, 2010, and 2020
- NHGIS attributes for population, housing, tenure, vacancy, income, race/ethnicity, and commute mode
- Harmonization to a common 2010 or 2020 geography
- Station buffers joined to census geographies and station opening years
- Direct zoning, land-use capacity, or pre-opening built-form measures around stations

P2 refinements

- Expand to other metros in Baum-Snow sample
- Use distance to nearest station (continuous)
- Historical station-siting instrument for causal test

Summary



P1 and P2 provide within-DFW equilibrium support for the model's municipal FOC.
P3 should be estimated as a station-area panel once the missing NHGIS files are added.